Viral Sample Pool Design and Deconvolution Project Progress Report

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1 Project objective and current progress

My goal is to implement the computational methods in *Pooling Strategy for Massive Viral Sequencing* and evaluate sample recovery using public HCV HVR1 sequences. The problem is to design overlapping pools that identify individual specimens, then assign the sequences observed in those pools back to their source specimens.

I have implemented the compatibility graph, Greedy Pool Design Algorithm (GPDA), tabu-search clique-cut selection, kGEM clustering, and generalized pool intersections and differences. I have also completed an exploratory evaluation using eight public specimens. The implementation is a chapter-based reproduction with documented interpretations of the printed pseudocode, rather than a claim of exact reproduction of the published experiments.

2 Methodology: public HCV dataset

2.1 Data source and specimen selection

I used the HCV HVR1 sequence deposit associated with Janiak et al. (2019), available on Zenodo under DOI 10.5281/zenodo.3266795 and the CC BY 4.0 license. The source contains reconstructed, error-corrected haplotypes and estimated frequencies in FASTA headers. These are not raw sequencing reads.

Before evaluating recovery, I selected the first eight files labeled chronic genotype 1 in numeric specimen-ID order: **53, 80, 142, 153, 157, 165, 200, and 201**. The rule is deterministic but does not produce a random or representative cohort. The filenames use genotype 1; the study describes its genotype-1 cohort as subtype 1b. I did not independently re-genotype the sequences.

Public specimen	Source haplotype records
53	16
80	70
142	35
153	24
157	20
165	5
200	54
201	114
Total	**338**

Preprint.

2.2 Preparation and pooling protocol

I retained all listed haplotypes without an additional frequency cutoff and normalized the header frequency estimates within each specimen. I removed the source gap symbols and aligned all sequences together using MAFFT 7.526 with `--auto --thread 1`. The alignment contains 180 columns and 333 distinct strings. Checksums verify the downloaded files, and sequence-content checks confirm that alignment did not change the ungapped nucleotide strings.

For this evaluation, I assumed all eight specimens were mutually compatible. Measured titers and epidemiological exclusions were not used. Thus, the experiment tests deconvolution under a complete compatibility graph, rather than validating the biological compatibility model.

I used a maximum pool size of four and independently sampled 80 observations per included specimen per pool using the normalized source frequencies. Each replicate has 1,040 observations. I ran five replicates using seeds 91 through 95. The initial clustering multiplier was W=2; deconvolution was repeated from the original pools with increased W, up to W=4, if any recovered specimen was empty.

The input haplotypes are real public HCV sequences, but pooling and read sampling are simulated. No laboratory pooling, new sequencing, or added sequencing-error model was used. Ground-truth source labels were withheld from clustering and used only to evaluate assignments. Independent observations retain distinct identifiers, even if their nucleotide strings are identical.

3 Algorithm

3.1 Pool design with GPDA and tabu search

Let G represent sample compatibility and H represent unresolved sample pairs. Initially, H contains every pair of specimens. A candidate pool A is valid if it is a clique in G and its size does not exceed the limit T=4. It separates a pair when exactly one endpoint belongs to A.

The chapter's clique-cut objective can be written as

$$f(A) = c_H(A, V \setminus A) - (|E(H)| + 1)q_G(A),$$

where c_H counts unresolved pairs separated by A and q_G counts incompatible pairs inside A. The penalty discourages invalid pools. The tabu heuristic searches both orientations of a MaxCut initialization, uses the chapter's single-vertex gain formulas to make improving moves, and uses undo operations and temporarily forbidden moves to explore alternatives. It trims an oversized record to satisfy T. The implementation also uses documented iteration limits and a singleton fallback.

GPDA adds the selected pool and removes the pairs it separates from H. It repeats until H has no edges, then adds an uncovered singleton if needed. Finally, it checks that every specimen is covered and has a distinct pool-membership signature.

For the public specimens, the resulting design was:

Pool	Specimens included
P1	53, 142, 157, 200
P2	53, 80, 157, 165
P3	53, 80, 142, 153
P4	201

This gives four pools for eight specimens. The same design is used across the five sampling replicates. GPDA and tabu search are heuristics; I do not claim that their output is globally optimal. The additional exact clique-cut solver is used only for small-instance verification.

3.2 kGEM clustering

For two pools, I set the requested cluster count to k=W times the number of specimens in their membership union. If k exceeds the number of distinct observed strings, the implementation caps it. The initial centroid is the most frequent read, and subsequent seeds maximize their minimum Hamming distance to the existing seeds.

For centroid g_i and aligned read r of length L, the emission likelihood is

$$h_{i,r} = \prod_{m=1}^{L} f_{i,m}(r_m).$$

The consensus allele at a position has probability 1-4 epsilon and each of the other four alleles has probability epsilon; the alphabet includes a deletion symbol. The current implementation uses epsilon=0.01. Given mixture proportions pi_i, posterior cluster membership is

$$p_{i,r} = \frac{\pi_i h_{i,r}}{\sum_j \pi_j h_{j,r}}.$$

The inner iteration estimates mixture proportions using observed read counts. The outer iteration updates the consensus alleles using the chapter's printed posterior sum over distinct read strings. This is a discrete consensus update, not a full soft allele-frequency update. Likelihoods are computed in logarithmic form for numerical stability. The iterations stop when estimates stabilize or a documented iteration or cycle limit is reached. Each observation is assigned to its highest-posterior cluster.

3.3 Generalized intersections and deconvolution

For pools P and Q, I cluster their combined observations and retain clusters that contain reads from both pools. These clusters form the generalized intersection. The known pool memberships determine which specimens this intersection represents; the nucleotide identities determine which reads enter it. Directed differences remove the selected observation IDs from the corresponding original pool.

The queue procedure repeatedly replaces a pair of overlapping pools by its intersection and nonempty directed differences until individual specimen sets remain. Replacing both operands is an explicit interpretation of the chapter's queue pseudocode. Increasing W is triggered only by an empty recovered specimen; a nonempty but incorrect result does not trigger another attempt.

4 Results on the public HCV dataset

Accuracy is the number of correct assignment entries divided by all assignment entries. Read recall is the number of correctly recovered observation IDs divided by all input observation IDs. Every observation was assigned exactly once in these runs, so accuracy and recall coincide.

Measure	Public HCV result
Specimens	8
Source haplotype records	338
Alignment columns	180
Pools	4
Observations per replicate	1,040
Mean assignment accuracy and read recall	71.81%
Accuracy range across five seeds	70.48–72.88%
Mean per-specimen recall (macro average)	72.97%
Mean recall excluding the singleton specimen	69.46%

4.1 Replicate results

Sampling seed	Correct assignments / observations	Accuracy	Final W
91	746 / 1,040	71.73%	3
92	758 / 1,040	72.88%	3
93	747 / 1,040	71.83%	4
94	750 / 1,040	72.12%	4
95	733 / 1,040	70.48%	3

The variation here comes from sampling the same underlying specimens repeatedly. It does not measure uncertainty across independent cohorts. All 48 kGEM fits in the main runs, including retries, reported inner and outer convergence. This shows numerical stabilization, not correctness of the resulting specimen assignments.

4.2 Specimen-level recovery

For seed 91, recovery differed substantially between specimens:

Public specimen	Correct / true observations	Recall
53	240 / 240	100.00%
80	6 / 160	3.75%
142	70 / 160	43.75%
153	80 / 80	100.00%
157	160 / 160	100.00%
165	50 / 80	62.50%
200	60 / 80	75.00%
201	80 / 80	100.00%

Specimen 201 occupies a singleton pool, so its complete recovery does not test separation of mixed populations. Excluding that specimen gives a more informative additional measure of performance on pooled specimens. Specimen 80 remains poorly recovered despite a nonempty output, showing why the empty-sample retry rule is not sufficient to detect failure.

4.3 Sensitivity to W

I also used seed 91 with retries disabled to examine fixed W values:

Fixed W	Correct / total observations	Accuracy	Empty recovered specimens
1	499 / 1,040	47.98%	80, 157
2	710 / 1,040	68.27%	80
3	746 / 1,040	71.73%	None

The results depend on W, but these three runs do not establish that increasing W always improves recovery. They are a sensitivity analysis on one sampling replicate, not a separate validation set or a parameter-optimization study.

5 Verification and limitations

The saved full test run contains 19 passing tests. Relevant checks include pool feasibility and separation, clique-cut gain formulas, source-file checksums, frequency parsing, preservation of ungapped sequence content, and metric accounting under deliberately incorrect assignments.

The main limitations of the present evaluation are:

- The dataset contains reconstructed haplotypes and frequency estimates, not raw reads or experimentally pooled specimens. The simulation does not include PCR bias, contamination, unequal titers, or sequencing errors.
- Only eight chronic genotype-1 specimens from one study were evaluated. All samples were assumed compatible, so the experiment does not test measured biological compatibility restrictions.
- One aligned sequence occurs in more than one specimen. Identical strings alone cannot uniquely establish the original source of an observation.
- Numerical convergence and nonempty recovered sets do not guarantee correct biological assignments. Poor recovery can remain undetected by the retry condition.
- The implementation includes documented repairs to stopping conditions, bounded search, and queue refinement. It is not a literal execution of every printed pseudocode line or a reproduction of the chapter's published empirical accuracy.
- Four pools instead of eight individually processed specimens demonstrates a reduction in pool count in this setup. It does not establish a 50% monetary saving, because total sequencing depth and laboratory costs were not held fixed.

6 Next steps

My next priority is to trace the generalized intersections that misassign observations from specimens 80 and 142. I plan to inspect the saved confusion matrices and clustering diagnostics, then vary W, sampling depth, and pool-size limits separately. I also plan to evaluate additional public specimens using selection rules set before inspecting their results. Any alternative consensus update or queue strategy will be identified as a departure from the current chapter-based implementation. These are planned investigations, not completed improvements.

7 References and supporting results

1. Skums et al. (2016). *Pooling Strategy for Massive Viral Sequencing*. Chapter 3 in *Computational Methods for Next Generation Sequencing Data Analysis*, pp. 57–83.
2. Janiak et al. (2019). *Hepatitis C virus (HCV) genotype 1b displays higher genetic variability of hypervariable region 1 (HVR1) than genotype 3*. Scientific Reports.
3. Caraballo Cortes, K. (2019). Public HCV HVR1 sequence dataset. Zenodo, DOI 10.5281/zenodo.3266795, CC BY 4.0.

- Public-data provenance and preparation
- Public HCV results and diagnostics
- Saved full test output